

SHERVEEN MEHRVARZAN

Master of Science, Mechanical Engineering, University of Washington

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EDUCATION

UNIVERSITY OF WASHINGTON

Master of Science, Mechanical Engineering

Bachelor of Science, Mechanical Engineering

- Mechanical Engineering Course Work: Kinematics & Dynamics, Mechanics of Materials, Material Science, Applied Computing, Engineering Statics, Chemistry Series, Calculus Series, Physics, Thermodynamics, System/Fluid Dynamics, Finite-Element Analysis, Embedded Systems, Machining

SEATTLE, WA

Graduating June, 2026

GPA 3.8/4.0 June, 2024

EXPERIENCE

AMAZON ROBOTICS

Electromechanical Co-op

SEATTLE, WA

January, 2025 — June, 2025

- Contributed to development and testing of the Transfer Cartesian Cell (TCC) machine for rapid tote transfer
- Spec'd out **pully mechanism** for bearing cart rail system, developed **in-house calculator for bearing life fatigue** under unique travel path
- Researched and developed **cost-effective mechanical gripper designs** to replace legacy pneumatic systems
- Achieved projected **cost savings of over \$3 million annually** per building through gripper redesign

SPACEX

Mechanical Engineering Internship

REDMOND, WA

June, 2024 — September, 2024

- Contributed towards top level design of **next generation antenna design**, focusing on main reflector geometry and optimization of antenna gain
- Rapid prototyping of 2 test enclosures for power electronics in high EMI environments, **reducing electrical interference** through EMC techniques
- Developed a **MATLAB script** that develops different sized antennas based off main/sub reflector diameter, subtend angle, and focus length
 - Formulated equations for aperture efficiency, feed losses, spillover efficiency, and geometry for a Gregorian style antenna
- Tested effects of environmental impacts on antenna attenuation and gain loss through **self-designed water and wind testing jigs**
- Redesigned AC units for gateway ground stations to withstand transportation vibration profiles by designing test brackets and vibration bed testing
 - Used **Mile's Equation and Miner's Rule** to solve for maximum G loads and fatigue stress to design brackets based off a PSD curve

SECOND ORDER EFFECTS

Electrical Engineering Internship

REDMOND, WA

September, 2023 — December, 2023

- Designing PCBs in **Altium** with custom circuitry, fault testing PCBs, and developing firmware on PCBs for companies
- Developed circuitry that is compatible for a high voltage and low voltage board
- Constructed DC load switching circuit with **current sensing** using ICs and AC current sensing with current shunts and amplifiers
- Read duty cycle from PWM signal through **active filtering** and buffering circuit

TESLA

Mechanical Engineering Internship

FREMONT, CA

June, 2023 — September, 2023

- Worked on Cybertruck **ultrasonic torque testing** for durability vehicles, collected clamp load data to help improve future joint designs
- Looked into Model 3 and Model Y O-ring failure in manifold by performing **resonance fatigue testing** on O-ring to conclude cause of failure
- Developed a test bench to **fatigue testing lateral links** for Model S/X/3/Y to solve a press fit failure issue with a bushing
- Helped design a brake dyno for future vehicle to test stick-slip phenomena, **designed adapters for half-shafts** with custom key and gear tooth
- Studied corrosion in areas of extreme weather on the subframes of the Model 3/Y and **implemented mitigation strategies to reduce corrosion**
- Planned out new test procedures to **validate new boot designs** for the knuckle to help water ingress and possible tearing

BOEING COMMERCIAL AIRPLANES

Product Development Engineering Internship

EVERETT, WA

June, 2022 — September, 2022

- Learned about **kinematic design** practices related to **wing edges mechanisms** on BCA planes
- Worked on **inboard flap torque tube** design for future BCA plane
- Designed an **inboard flap mechanism** for as a capstone project in a kinematics class
- Took a class on **carbon fiber layups** and bagging procedures at the Boeing Concept Center

UW FORMULA MOTORSPORTS TEAM

Electric Powertrain Lead

SEATTLE, WA

June, 2024 — June, 2025

- Designing a **537.5V battery with a 128S1P design**, completely new redesign of the high voltage battery for the team
- Setting performance goals out of the high voltage system and understanding **thermal losses and efficiency**
 - Developed in-house simulations using Python to recreate **battery thermal characteristics with active cooling** under charging and discharging conditions
- Taught younger engineers** about first principal concepts in both mechanical and electrical engineering
- Managing the safety and circuitry behind the functionality of the high voltage battery

Firmware Member

June, 2023 — June, 2024

- Developed firmware for charging a 550V battery pack using **Python to read and write CAN messages**
- Created a GUI for a user to charge the battery pack and created a **state-machine** for security and safety checks as the pack charges
- Helping out the development of a Analog-to-CAN circuit board by debugging ADC and CAN transceiver through oscilloscope SPI lines
- Wrote a C++ library** for an LTC2449 ADC for reading analog sensors across the car
- Managed 30+ sensors across the car through parsing analog data from 5Hz to 200Hz and delivering them as CAN messages

Technical Director

June, 2022 — June, 2023

- Overseeing all technical operations on the team of 80+ members, reviewing both mechanical and electrical designs
- Involvement in **integration of electrical and mechanical systems** of the vehicle
- Constructing top level design goals and organizing a timeline for technical projects and teams
- Understanding of 4WD car vehicle dynamics and EV control systems design, battery management, composite layups, drive train mechanics
- Placed **3rd overall** against 73+ vehicles at the **FSAE Michigan 2023** competition, first UWFM electric vehicle to participate in all dynamic events
 - 3rd Acceleration, 3rd AutoX, 2nd Endurance, 5th Design, 5th Skidpad

Suspension Lead

March, 2021 — June, 2022

- Teaching kinematic suspension geometry, vehicle dynamics, mechanical design, integrating vehicle performance with mechanical design
- Validating structural design through **mechanics of materials** concepts and **finite element analysis simulations in Ansys**
- Managing advanced assemblies and crafting difficult parts in **SolidWorks**
- Manufacturing parts by writing **CAM for CNC 2-Axis Mills** and Conventional Lathes